ANALYSIS REPORT · SLITTER 2

Out of an 8-hour day, Slitter 2 ran for only 1 h 51 min

Day analysed: Monday, 12 October 2026 · Sources: Slitter 2 PLC log, job records (ERP), CCTV footage (11:24–12:02)



11:42 — The next coil is already hanging on the crane, but no one is near the line. This lasted about 14 minutes (11:37–11:50).

23%	373 min	13	152 m/min
Utilisation 111 of 484 working min	Time stopped (excluding lunch)	Coils processed	Speed when running Stable (not the issue)

The conclusion in one line

The machine is not slow — it stands still too often and for too long.

- **Biggest loss: same-spec coil changes (184 min).** No knife change was needed, yet the line stopped 20 min for every coil, because unloading, banding and labelling are all done with the line stopped.
- **Spec changeovers took twice the usual time (48 vs 22 min).** On video, no one moved for 14 min after the next coil was hung; knife work started 30 min after the line stopped.
- **Fixing this could recover about 2.5 hours a day (estimate).** Most of it means changing the **order of work**, not spending money.

0 Problems and solutions at a glance

What is wrong

#	Problem	Loss today
1	**Same-spec coil changes stop the line for 20 min each**, even with no knife change. Unloading, banding by destination and labelling are all done with the line stopped.	**184 min** largest
2	**Spec changeovers took twice the usual time (48 vs 22 min).** On video, no one was at the line for 14 min after the next coil was hung; knife work started 30 min after the stop.	**143 min**
3	**27 minutes from power-on to the first coil** — repeated every morning.	**27 min**
4	**ERP start/finish buttons are pressed late or skipped**, so ERP times cannot show whether improvements work.	—

Not the problem: **speed**. When the line runs it holds 152 m/min.

What to do — run-time work stays in run time

#	Action	How	Effect / day (estimate)
1	**Move banding & labelling off the line**	Tie a temporary band, unload at once; do final banding while the next coil runs.	**~70 min**
2	**Pre-assemble the knife set**	Build the next knife/spacer set while the current coil runs; when the line stops, start with the knives.	**~70 min**
3	**Prepare the next coil and labels while running**	Hang the next coil by crane and print labels before the current coil ends.	2–3 min / coil
4	**Set up tomorrow's first job today**	Before leaving, set knives, spacers and first coil for the next morning.	**~15 min**
5	**Fix roles during changeovers**	Knife lead / next-set prep / unloading & banding.	makes 1–3 stick
6	**Press buttons on time + add stop-reason buttons**	Record start/finish when they happen; tag every stop with a reason.	makes it measurable

Expected result (estimate, based on today)

About 2.5 hours back per day

Utilisation 23% → about 35%. Most of it is changing the **order of work**, not buying equipment.

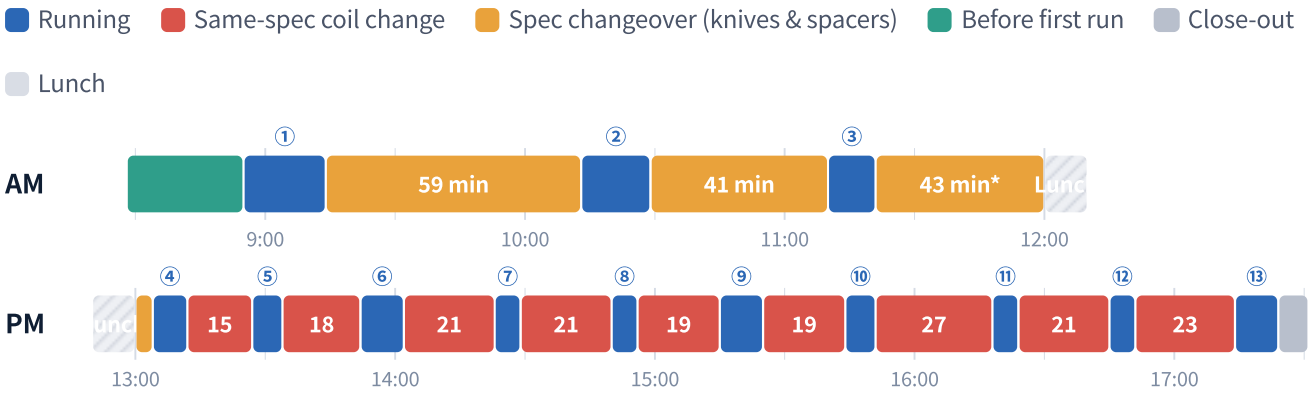
Start here

This week: fix changeover roles (#5) and the rule "when the line stops, knives first".

Next week: pick one day with a run of same-spec jobs and trial off-line banding (#1).

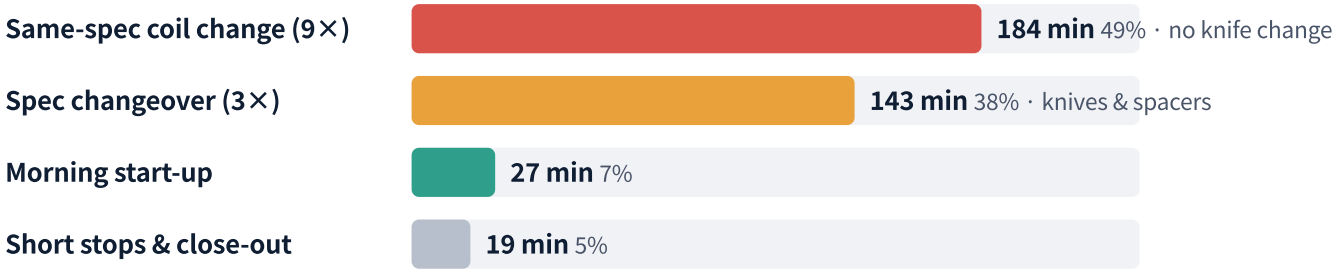
1 What happened during the day

The bars show, in time order, when Slitter 2 was **running (blue)** and when it was **stopped (colour = reason)**. Circled numbers are coil order; numbers inside bars are minutes stopped.



* The third spec changeover (11:21–13:04) is counted as 43 min after removing the 60-min lunch. Thin blue bars = short runs, long stops.

Where did the 373 stopped minutes go?



Put simply

In the morning the time went into **changing specs**; in the afternoon, into **swapping identical coils**. The 9 afternoon coils had exactly the same spec — no knife change — yet each one stopped the line for about 20 min.

Today's job mix

AM: 3 different specs (0.40, 0.77, …)
PM: Yuchang 1.15X4XC (57=19 Sihwa + 131=1 Hwaseong, banded separately), **10 in a row**
Coil length: AM ~1,500–2,400 m, PM ~900 m



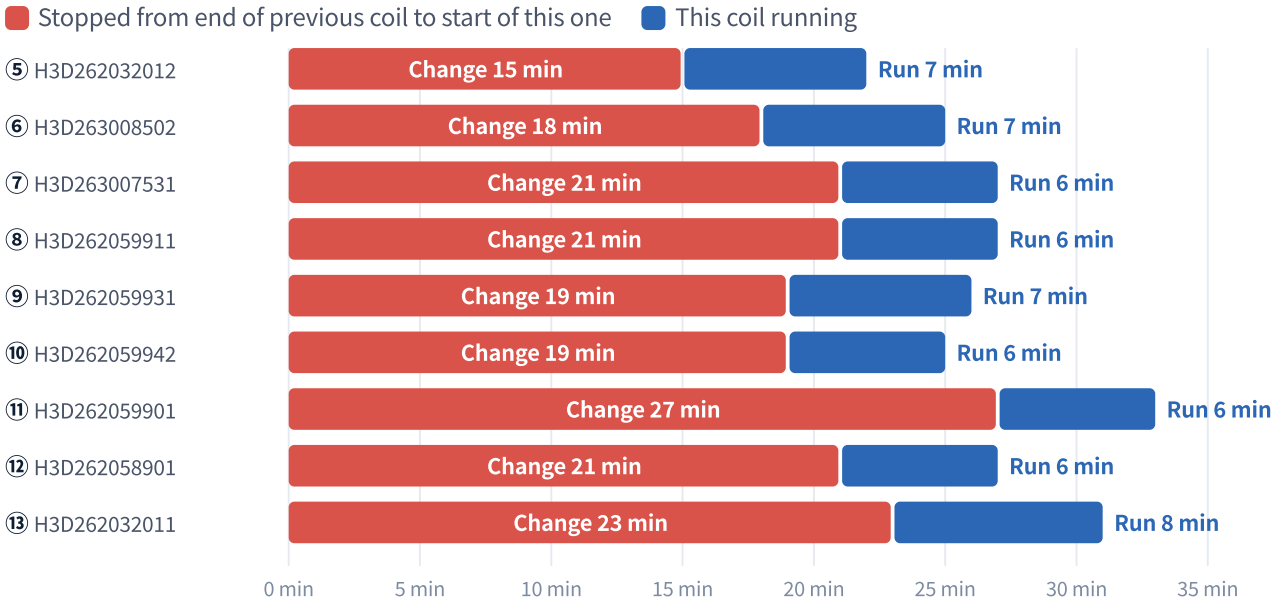
11:31 — Changeover in progress: several people moving between the head and the panel. The line is stopped throughout.



11:57 — Finishing loading the next coil, but lunch is about to start; this coil only started running at 13:04.

2 Problem ① 20-min stop per coil, with no knife change

The 9 afternoon coils had **identical specs and slitting patterns**. No knives or spacers needed changing, yet each coil **ran for 6–8 min** and **stood for 15–27 min.**



Why so long? (likely)

With no knife change, the 20 min must go on:

- Unloading 20 slit coils from the recoiler
- **Separate banding** per destination (Sihwa 19 / Hwaseong 1)
- Labels and markings
- Hanging and threading the next coil

All of this is done with the line stopped.

Put simply

Each coil takes 27 min, but the machine works for only **a quarter of it (6.5 min)**. A same-spec change usually takes 13.5 min (6-week median), so today each one ran about 7 min over.

9 × 20 min = 184 min
half of all stopped time



11:25 — Clearing the previous job (800421) at the recoiler/panel side. The line is stopped meanwhile.



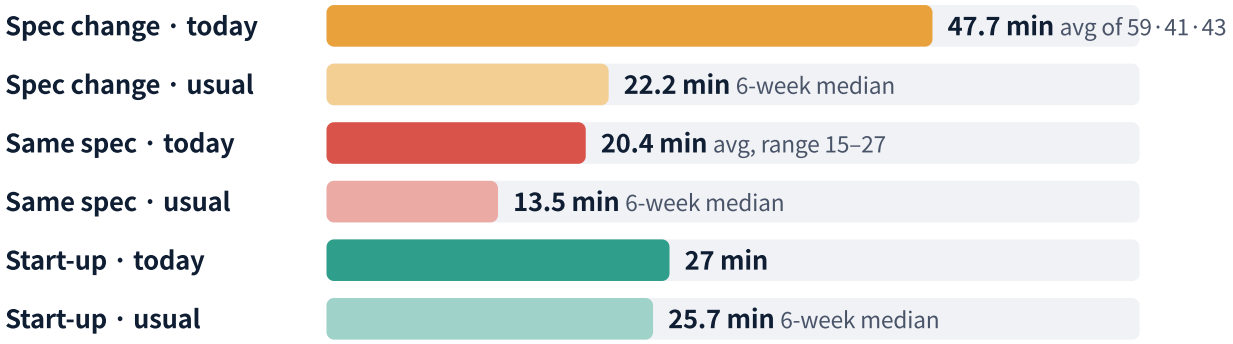
11:28 — Two people handling product at the recoiler. In the afternoon this unloading and banding was repeated for every coil.

Note: afternoon footage has not been received yet, so these photos are from the morning. With the 13:00–17:30 video, unloading, banding and loading can each be timed exactly.

3 Problem ② Spec changeovers took twice the usual time

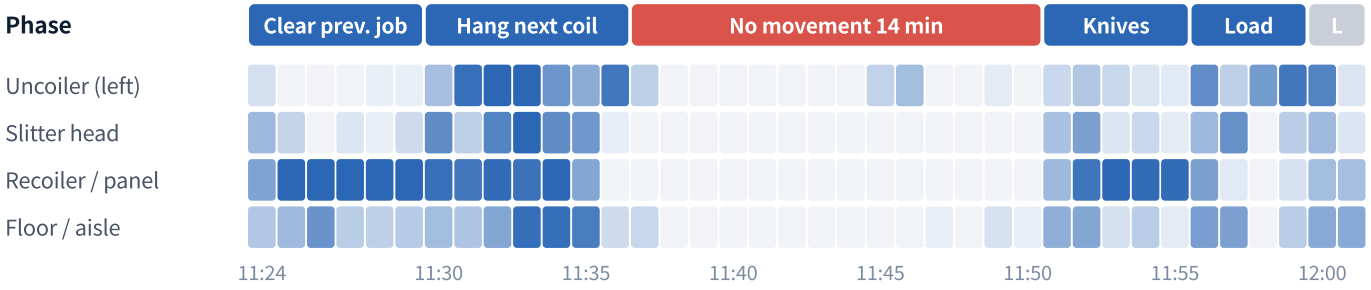
There were three knife/spacer changeovers: **09:14–10:13 (59 min)**, **10:29–11:10 (41 min)** and **11:21–13:04 after lunch (43 min)**. The average of 48 min is twice the 6-week norm (22 min).

Today vs usual (average per change, minutes)



The third changeover on video (11:24–12:02)

We compared the CCTV image every second and coloured **how long people or objects moved in each zone**. Darker = more movement in that minute; grey = no movement at all (L = lunch).



What stands out

- **11:37–11:50: zero movement in all 4 zones for 14 min.** The next coil was already hanging on the crane.
- Knife/recoiler work only began **at 11:51** — 30 min after the line stopped.
- Of 37 min, 23 had movement and **14 (38%) had none**.

To be confirmed

The ERP start and coil number were entered at 11:47, so someone was working **off-camera**. Was this 14 min (1) spacer work out of view, (2) plain waiting, or (3) slowing down before lunch? The spacer bench is not visible to the camera.



11:55 — Knife work at the head, 34 min after the stop.



11:59 — Coil still hanging as lunch begins.

3 Problem ② Step by step — changeover 11:24–12:02

Same camera, same angle. Left: uncoiler (where the coil is hung). Centre: slitter head. Right, behind the yellow rail: control panel and recoiler.



① **11:33** — Several people moving around uncoiler, head and panel: clearing the last job and preparing the next coil.



② **11:36** — The next coil (H3D262059952) is hung on the crane. **Everything on track so far.**



③ **11:42** — **No one there.** The coil waits on the crane.



④ **11:47** — **Still no one.** ERP start was pressed at this moment, but nobody is in view.



⑤ **11:52** — Work resumes: knife/arbor work at the head and recoiler (31 min after the stop).



⑥ **11:57** — Coil loaded and jogged. But lunch starts at 12:00, so the line actually runs at 13:04.

Put simply

Between ② hanging the next coil and ⑤ starting the knives, **15 minutes were empty.** The real knife and coil work was then squeezed in just before lunch, so the line only ran after lunch at 13:04. Simply following **"when the line stops, knives first"** would have had this coil running before lunch.

4 Other problems

Problem ③ 27 minutes to the first run

The PLC powered up at **08:28** and the first coil started at **08:55**. Over the last 6 weeks the first job of the day also took about 26 min to prepare — **a loss repeated every day**. If the shift starts at 08:00, the real loss is bigger.

> **Direction**
> After the last job of the day, **set the knives, spacers and first coil for tomorrow's first job**. In the morning, just thread and run.

Problem ④ Job records do not match reality

Item	Actual (PLC)	Recorded (ERP)
First coil (2,369 m)	Ran 08:55–09:13	**No coil number, no job record**
H3D263008502	Ran 13:52–14:01	Start 14:03:21 → finish 14:03:23 **(2 seconds)**
H3D262059952	Ran 13:04–13:11	Start 11:46 → finish 13:14 (87 min incl. lunch)

Start/finish buttons are sometimes **pressed in batches afterwards or skipped**. ERP times then cannot measure work time or show whether improvements work, so this report is based on the **PLC (the machine's own log)**.

What is not the problem

- **Speed**: always 150+ m/min when running. With only 23% running time, faster speed would save just a few minutes.
- **Preparing the next coil**: hanging it in advance by crane is already done. But it happens **after** the line stops, so it can be brought forward.



11:51 — Work resumes at the head. Knife work started from this point.



12:01 — Lunch. The coil is hung, but the line stays stopped until 13:04.

5 Improvement plan

One principle: **"Work that can be done while the line runs is done while the line runs."** Today most unloading, banding and knife preparation happens with the line stopped.

#	What	How (simply)	Expected effect (estimate, today)
1	**Move banding & labelling off the line**	At the recoiler, apply only a temporary band and unload at once. Do final banding by destination and labelling while the next coil runs. Print labels before the job.	**~70 min/day** 20→12 min × 9
2	**Spec change: pre-assemble the knife set**	While the current coil runs, build the next knife/spacer set on a bench. When the line stops, **start with the knives**.	**~70 min/day** 48→25 min × 3
3	**Hang the next coil while running**	During the current coil, crane the next coil next to the uncoiler. Prepare labels and job sheets at the same time.	within 1 & 2 2–3 min each
4	**Set up tomorrow's first job today**	Before leaving, set knives, spacers and first coil. In the morning, thread and run.	**~15 min/day** 27→~10 min
5	**Fix roles during changeovers**	Operator = knives & threading; spacer role = next set; helper = unloading & banding; shared staff = crane & labels.	keeps 1–3 in place
6	**Buttons on time + stop reasons**	Press start/finish when it actually happens. Add **stop-reason buttons** on the floor screen (unload / band / knives / waiting / material).	makes effect measurable

Expected result (estimate, a day like today)

About 2.5 hours back per day

Utilisation 23% → about 35%. Enough to finish earlier or run **5–8 more** of the afternoon 1.15 mm coils. Simply getting back to the **usual level (6-week median)** recovers about 2 hours.



What already works — 11:36, the next coil hung in advance. Doing this **while the current coil runs** is action #3.

Action #1 as a picture (afternoon same-spec coil, example)

Now

Unload | Band & label | Hang coil | Thread | Run 6.5 min

Stopped ~20 min → ~27 min per coil

After

Unload+tie | Load | Thread | Run 6.5 min

Band, label, next coil ← while the line runs

Stopped ~12 min → ~18 min per coil

Step times are illustrative; they will be replaced with measured times once the 13:00–17:30 footage is available.

6 Roll-out and open questions

When	Action	Cost
This week	① Fix changeover roles (#5) ② Rule: "when the line stops, knives first" ③ Press buttons on time (first half of #6)	None
Next week	④ On a day with many same-spec jobs, **trial off-line banding for one day** (#1) — check unloading cart and banding space ⑤ Start next-day set-up (#4)	None–small
Within 2 weeks	⑥ Secure a bench for pre-assembling knife/spacer sets (#2) ⑦ Add stop-reason buttons to the floor screen (#6)	Small
After 1 month	⑧ Re-measure utilisation and changeover times from the PLC and compare with today (23%, same-spec 20 min, spec change 48 min)	None

Questions for the shop floor

- 11:37–11:50 (14 min): what was the crew doing while no one was at the line? (spacer assembly / waiting / other work)
- Is **separate banding** of the 20 strips by destination a customer requirement? Is there space and equipment (unloading cart, stands) to do it off the line?
- What time does the shift start? (If 08:00, the morning loss is more than 27 min.)
- Can knife/spacer sets be assembled next to the line in advance? (spare arbor or sets?)

Footage still needed

- **12 Oct 13:00–17:30** (most important) — 9 same-spec changes; lets us time unloading, banding and loading step by step.
- **12 Oct 08:30–11:24** — the 59-min and 41-min spec changeovers.
- Next time, also record **a camera covering the spacer bench** — to tell "waiting" from "off-camera work".

Data and method

• Running/stopped: Slitter 2 PLC log (~2.3 s interval) averaged per minute; **average speed ≥ 30 m/min = running**. 08:28–17:31 = 544 min; working time 484 min excluding lunch 12:00–13:00.
• Usual level: median stop time per coil for 122 Slitter 2 PLC coils, 1 Sep – 9 Oct 2026 (same spec 13.5 min / spec change 22.2 min / first job 25.7 min).
• Job content: specs and slitting patterns from the 11 Slitter 2 ERP job records. Their start/finish times differ from reality and were not used for timing.
• Video: CCTV 11:24:24–12:02:07 (37 min) compared every second; movement in 4 zones judged from the share of pixels changing over 2 s. Crane motion also counts as movement; work outside the camera view is not captured.
• Expected effects are **estimates** based on this one day and will vary with the job mix (number of spec changes, strips, coil length).

Appendix Coil-by-coil record (PLC)

#	Running	Coil no.	Spec · pattern	Length (m)	Run (min)	Stop before (min)
①	08:55–09:13	(no number)	-	2,369	17	Start 27
②	10:13–10:28	T262205557302	0.40 · 77=15, 60=1	2,403	16	Spec 59
③	11:10–11:20	800421	0.77 · 162=5 etc., trim & special banding	1,465	11	Spec 41
④	13:04–13:11	H3D262059952	1.15 · 57=19, 131=1	902	8	Spec 43*
⑤	13:27–13:33	H3D262032012	same	906	7	Same 15
⑥	13:52–14:01	H3D263008502	same	906	7	Same 18
⑦	14:23–14:28	H3D263007531	same	901	6	Same 21
⑧	14:50–14:55	H3D262059911	same	891	6	Same 21
⑨	15:15–15:24	H3D262059931	same	891	7	Same 19
⑩	15:44–15:50	H3D262059942	same	890	6	Same 19
⑪	16:18–16:23	H3D262059901	same	893	6	Same 27
⑫	16:45–16:50	H3D262058901	same	889	6	Same 21
⑬	17:14–17:23	H3D262032011	same	884	8	Same 23
Total					**111**	**354** + short stops 11 + close 8

* Lunch (60 min) excluded. "Start" = power-on to first run; "Spec" = changeover with knife/spacer change; "Same" = change between coils of the same spec. Short stops of 1–3 min during a coil (11 min in total) are counted separately. Coils ⑤–⑬ are all Yuchang 1.15X4XC, 57=19 (Sihwa) · 131=1 (Hwaseong), banded separately.